

PROGRAM

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#include <stdio.h>

#include <stdlib.h>

// Node structure definition

struct Node {

    int data;

    struct Node* next;

};

// Function prototypes

struct Node* createNode(int data);

void push(struct Node** top, int data);

int pop(struct Node** top);

void displayStack(struct Node* top);

int main() {

    struct Node* top = NULL;

    int choice, element;

    while (1) {

        printf("\nStack Operations Menu:\n");

        printf("1. Push\n");

        printf("2. Pop\n");

        printf("3. Display\n");

        printf("4. Exit\n");

        printf("Enter your choice: ");

        scanf("%d", &choice);

        switch (choice) {

            case 1:
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        printf("Enter element to push: ");

        scanf("%d", &element);

        push(&top, element);

        break;

    case 2:

        element = pop(&top);

        if (element != -1)

            printf("Popped element: %d\n", element);

        break;

    case 3:

        displayStack(top);

        break;

    case 4:

        exit(0);

    default:

        printf("Invalid choice! Please enter a valid option.\n");

    }

}

return 0;

}

// Function to create a new node

struct Node* createNode(int data) {

    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));

    if (!newNode) {

        printf("Memory allocation error\n");

        exit(1);

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    }

    newNode->data = data;

    newNode->next = NULL;

    return newNode;
}

// Function to push an element onto the stack
void push(struct Node** top, int data) {

    struct Node* newNode = createNode(data);

    newNode->next = *top;

    *top = newNode;

    printf("Element pushed: %d\n", data);
}

// Function to pop an element from the stack
int pop(struct Node** top) {

    if (*top == NULL) {

        printf("Error: Stack underflow. Cannot pop element.\n");

        return -1;

    }    struct Node* temp = *top;

    int poppedElement = temp->data;

    *top = (*top)->next;

    free(temp);

    return poppedElement;
}

// Function to display the stack elements
void displayStack(struct Node* top) {

    if (top == NULL) {

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        printf("Stack is empty.\n");
        return;
    }
    printf("Stack elements: ");
    struct Node* temp = top;
    while (temp != NULL) {
        printf("%d->", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
```

OUTPUT:

Stack Operations Menu:

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter element to push: 10

Element pushed: 10

Stack Operations Menu:

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter element to push: 20

Element pushed: 20

Stack Operations

Menu:

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter element to push: 30

Element pushed: 30

Stack Operations Menu:

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

Stack elements: 30 20 10

Stack Operations Menu:

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 2

Popped element: 30

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 3

Stack elements: 20 10

Stack Operations Menu:

1. Push

2. Pop

3. Display

4. Exit

Enter your choice: 4